

# How Would You Like Your Recommendations? p-book, a Self-Recommending Living Book That Teaches How Recommender Systems Work

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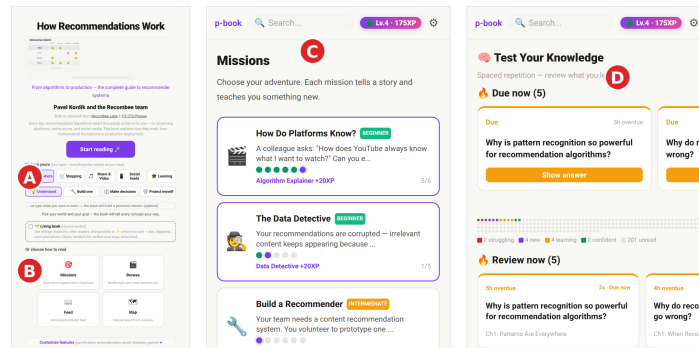


Figure 1: Surfaces of p-book: home screen with onboarding picker (A) and paradigm tiles (B); story-driven Missions (C); spaced-repetition review (D). Live: <https://recsys-pbook.vercel.app>.

## Abstract

A recommendation platform decides three things for its users: how content is *served*, how each item is *told*, and what *exists* in the catalog at all. We demonstrate **p-book**, an open-source living book about recommender systems that hands all three decisions to the reader as explicit, logged, reversible choices—while a production recommender (Recombee) personalizes the book itself, so readers watch the mechanisms they are reading about act on their own behavior. Content is organized as human-contracted *concepts* told through facet-tagged *tellings*; readers steer any section, are answered from the curated catalog first, and may fill an honestly reported gap by generating or remixing a telling that then climbs a provenance-preserving curation ladder. The result is an *elastic* catalog—items on demand with curation-as-a-service—and an instrumented, editable research deployment that reviewers can use live.

## CCS Concepts

• Information systems → Recommender systems; • Human-centered computing → Interactive systems and tools.

## Keywords

recommender systems, user control, generative content, elastic catalogs, human-AI co-creation, algorithmic literacy

## 1 Introduction

Recommender systems shape most of what users encounter online, yet three of their defining decisions are never offered to the user: the consumption *paradigm* (an infinite feed, a grid of carousels,

a guided sequence), the *form* in which each item is told, and the *composition* of the catalog itself. User-control research has progressed *within* this frame—sliders, algorithm choice, explanatory interfaces [6–8]—and adaptive hypermedia has long adapted *presentation* [1, 2]; the frame itself, where most of a platform’s power resides, stayed fixed. We demonstrate **p-book**, a deployed system that inverts this default at all three levels. The reader chooses the consumption paradigm and may switch anytime; steers how any section is *told*—gentler or deeper, textual or visual, in their own world and language; and, when no telling fits, *creates* one: generation and editing are first-class, governed interactions. Treating autonomy as a design principle rather than a settings option [10], every exercised choice is a logged event under randomized initial assignment, so the deployment is at once a product and a research instrument.

The first p-book, *How Recommendations Work*, makes the instrument self-referential: a book that teaches how recommender systems work, with tellings written for readers from school age (a design target [5], not yet a validated one) up to research-level treatments, personalized by a production recommender, so the dwell, rating, and steering signals a reader emits visibly reshape what it suggests next [4]. We are not aware of another deployed system offering paradigm choice, telling-level steering, and catalog participation together; the claimed novelty is this integration—and a recommender demo that explains itself *with* itself. It contributes: (1) consumption-paradigm choice as a logged, randomized experimental variable; (2) a content model separating human-owned concept contracts from facet-tagged tellings with declared subspace coverage; (3) governed reader co-creation—segment-scoped

generation, remixing, and a provenance-preserving curation ladder; and (4) an open, instrumented deployment to study and edit.

## 2 A Living Book

**Concepts, contracts, and tellings.** The corpus separates *what must be learned* from *how it is told*. A *concept* is a unit of understanding with a human-approved, machine-readable *contract*: a learning objective, must-cover points, one canonical recall question with its answer, and forbidden claims; every generated or adopted telling is checked against it. A *telling* is one presentation of a concept, tagged on controlled facet dimensions—example world, depth, visibility, formalism, length, genre, and language (Table 1). The contract fixes *what* every telling must teach; the facets—a controlled vocabulary doubling as the generation interface—describe *how*; the recommender chooses only among tellings satisfying the contract. A telling declares the *subspace* it covers, may serve several concepts, and *covers* a requested point rather than equals it. Facets sync to the production recommender as item properties, so the same engine that ranks tellings learns per-reader facet affinities, shown to the reader as an editable open learner model [3] that explicit choices always override [9].

**Steering, honest misses, serve-or-mint.** The policy is *retrieve before generate*. The reader steers any section from a panel showing every existing telling with its provenance; requests *compose*, profile-seeded and overridable per dimension (genre *story* + language *Czech* + world *e-commerce*), the composed target visible before anything is served. When nothing qualifies, the miss is reported *honestly*—“no such telling yet”—never silently substituted; the gain is trust and a new signal class: every honest miss is a logged, rankable statement of demand for an item that does not exist. Only at a real gap, on an explicit action, is a telling *minted* from an auditable prompt (contract, facet vector, voice exemplar, correction ledger, existing titles, and the reader’s capped wish). It must pass a deterministic gate—must-cover coverage, formalism and length lints, tags matching what the output actually carries—before it is shown; human review happens later, on the curation ladder. Generation is *multi-modal* within one design system: the genres *comic* and *animation* yield real SVG tellings, and text tellings requesting visual carriers arrive with a drawn, validated diagram attached—so a request is never silently downgraded to prose. Unlike personalized generation aimed at one user [11, 12], it is *segment-scoped*: written for everyone who shares the requested facets, cached by facet vector, never impersonating curated content; personalization happens in *selection*. Readers improve any section through one entry point offering a manual edit or an AI rewrite; both yield a highlighted preview and, on accept, a persistent private version—original untouched, intent kept as provenance.

**Curation as a service.** Co-created content earns trust in stages, each an explicit item state serving conditions on: creations start private; sharing requires consent (anonymous by default) into a labeled community layer; the ladder—community, edited, core—climbs only through human review (Fig. 2). The tiered certificate is computed over human-verified core content only, so co-creation grows the catalog without lowering the certified bar. Editing is *demand-driven*: the same engine that ranks tellings for readers ranks *work for editors*—community tellings nominated once enough readers engage,

honest-miss logs as a ranked “asked for and not served” list, and per-telling health signals with one-action remedies; repeated fixes become rules in the correction ledger fed to every later generation. Every creation is durably archived with its wishes and diffs; adoption is one action—flipping the ladder state instantly changes what serving delivers to everyone, git remaining the canonical home of core content. Readers can even propose missing *concepts*: the system drafts proposals (contract included) from unserved demand and recommends them as *proposed*, *not written yet* cards, so votes measure demand before anyone writes.

## 3 What Changes with an Elastic Catalog

When items are created on demand [13] and trust is assigned after publication, several fixed assumptions stop holding. The catalog becomes a grid of *potential* items—every (concept × facet-subspace) cell could exist—and answering a request becomes a serve-or-mint decision. Minted items are *metadata-first*—born with dense features and a known requester segment—so cold start becomes every item’s birth state. Minting authority is a spectrum—reader (p-book today), editor, or a system pre-generating predicted demand; at the far end an unbounded generator drives exploration budgets toward zero. p-book therefore mints on demand and limits supply—serve first, cache identical requests, cap and retire unused cells, exploration as an explicit budget; the conservative *reuse threshold* is the essential tunable.

## 4 The Demonstration

The demo runs in any browser; reading works offline, generation and live recommendations need the network. Every step runs live—nothing is prerecorded.<sup>1</sup> A three-minute loop walks the control ladder:

- *serving* — flip the same content among quests, shelves, feed, and map; pin a preference (Czech · visual) and watch shelves reorder;
- *telling* — flip one concept (*item cold start*) through its tellings (Fig. 2), comic to Czech e-shop story to formal treatment (each concept ships a comic); steer to an uncovered cell and receive the honest miss;
- *catalog* — mint the missing telling (animated, if asked), edit a sentence by hand, remix another with a prompt, share it, and adopt it in the editorial queue—one click later it serves to every reader;
- *loop closed* — read the implicit-feedback section and watch just-emitted signals reshape the transparent profile; the examiner meets a vague answer with a probing follow-up, not a keyword score.

Reviewers drive every step from the live link; /admin shows the editor queues. Attendee creations stay in the community layer (promotion is an editor action); profiles are local, pseudonymous, erasable.

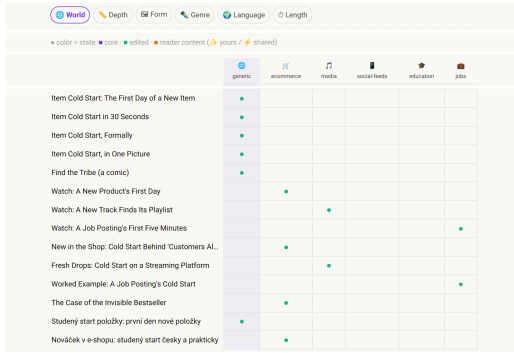
## 5 Outlook

The book is a running experiment. *Can the interface itself be personalized?* First-release telemetry: 42 readers, 177 voluntary paradigm switches; the open logs invite study. *Can LLMs produce content worth curating?* Come edit the book.

<sup>1</sup>Live: <https://recsys-pbook.vercel.app> · source: <https://github.com/kordikp/recsys-pbook> · a video walkthrough accompanies the submission; links are not anonymized, as permitted for demos.

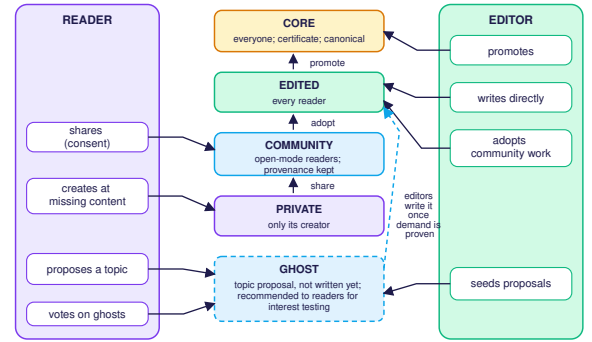
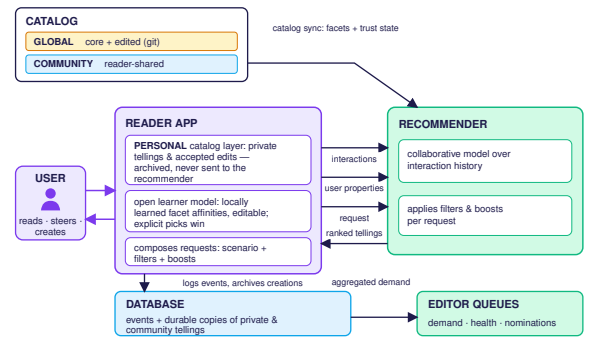
**Table 1: The telling taxonomy. Bold = implemented in the demo; plain = possible extensions. Supplementary to §2.**

| Dimension   | What it captures                                        | Values                                                                                                                                                                    |
|-------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lens        | the world all examples are drawn from                   | <b>generic</b> · <b>e-commerce</b> · <b>media</b> · <b>social feeds</b> · <b>education</b> · <b>job boards</b> · news · travel · real estate · fashion · gaming · finance |
| depth       | assumed reader background                               | <b>intro</b> · <b>standard</b> · <b>technical</b> · <b>research</b>                                                                                                       |
| visuality   | how much visuals carry the point                        | <b>text-first</b> · <b>balanced</b> · <b>visual-first</b>                                                                                                                 |
| formalism   | amount of mathematical notation                         | <b>none</b> · <b>light</b> · <b>full</b>                                                                                                                                  |
| length      | reading commitment                                      | <b>tl;dr</b> · <b>standard</b> · <b>deep</b> · deep w/ tl;dr lead                                                                                                         |
| genre       | structural form                                         | <b>explainer</b> · <b>story</b> · <b>worked example</b> · <b>code walkthrough</b> · <b>comic</b> · <b>animation</b> · quiz · cheat sheet · slides · FAQ                   |
| language    | natural language                                        | <b>English</b> · <b>Czech</b> · German · Spanish · French · Chinese                                                                                                       |
| carriers    | building blocks present; derived from content           | <b>prose</b> · <b>table</b> · <b>diagram</b> · <b>image</b> · <b>animation</b> · <b>formula</b> · <b>code</b>                                                             |
| trust state | provenance rung; gates serving, labeling, certification | <b>private</b> → <b>community</b> → <b>edited</b> → <b>core</b>                                                                                                           |

**Figure 2: The living-book map: rows are tellings, marked cells their declared subspace under the selected dimension (here: example world)—the *item cold start* concept across worlds and languages. Supplementary to §2 and §4.**

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**Figure 3: The content ladder with its zeroth rung: *ghost* topics are recommended for interest testing before anyone writes; readers create from below (private → community), editors write directly into *edited/core* and promote by review. Supplementary to §2.****Figure 4: The personalization loop as deployed: the reader app holds the personal catalog layer and the editable open learner model; global and community layers sync to the recommender, which returns ranked tellings; events and creations archive to a separate store feeding the editors’ queues. Supplementary to §2.**